

SOFTWARE PROJECT MANAGEMENT

INVENTORY MANAGEMENT SYSTEM (IMS) COMPLETE

Group Members:

1. MUTANGANA Joseph 29061
2. Musangwa Nshimiye Bennett 29092
3. Nshimyumuremyi Elie 26476
4. Shema Abdunur 28334



Contents

Inventory Management System (IMS)	2
Needs Assessment & Business Case	2
Identifying and Engaging Stakeholders.....	5
Forming The Team	6
RACI Chart	8
TEAM CHARTER.....	8
GROUND RULES	9
PROJECT SCOPE MANAGEMENT	10
REQUIREMENTS	10
MoSCoW PRIORITIZATION	12
ACCEPTANCE CRITERIA	12
DELIVERABLES	12
OUT OF SCOPE	13
CONSTRAINTS.....	13
ASSUMPTIONS	13
Working Breakdown Structure & Work Packages	13
Working Breakdown Dictionary.....	14
Activity Dependencies	16
Time Estimation	16
Resource Calendar	18
Risks	19
Negative Risks	19
Positive Risks	19

Inventory Management System (IMS)

Needs Assessment & Business Case

1. **Introduction / Background:** The Inventory Management System (IMS) project is planned to fix problems in how businesses manage stock, track products, and keep inventory records. Many companies still use manual methods or separate systems, which can cause mistakes, delays, and higher costs. IMS will give a single, reliable solution for managing inventory.
2. **Business Goals**
 - Make inventory records more accurate
 - Reduce stock shortages and extra stock
 - Improve work efficiency
 - Help make better decisions with up-to-date inventory information
3. **Current Situation and problem/opportunity statement:** Tracking inventory is slow and often has mistakes because records are kept by hand and not updated in real time. This is a chance to create an automated system that saves time, reduces losses, and makes customers happier.
4. **Critical Assumptions and Constraints**
 - Users know basic computer skills
 - Necessary computers and internet are available
 - Management supports the project

Limits(Constraints):

- Small project budget
- Fixed timeline
- Small development team

5. Analysis Options and Recommendation

- Option 1: Keep using manual inventory (cheap but risky)
- Option 2: Buy ready-made software (medium cost, less flexible)
- Option 3: Build a custom IMS (more work, best fit)

Recommendation: Option 3 is recommended because it meets the business needs and can be customized.

6. Preliminary Project Requirements

- Manage products and stock
 - Record stock coming in and going out
 - Send alerts for low stock
 - Provide simple reports
7. **Budget Estimate:** The estimated cost covers software development, testing, and launching the system. The cost is moderate, good for a small-to-medium project.
8. **Timeline Estimate:** The project should take about 3–4 months, including analysis, design, development, testing, and launch.

Project Timeline (3-4 Months)

Month 1 – Planning & Requirements

- Project Charter
- Stakeholder Identification
- Requirements Gathering & Approval

Month 2 – System Design

- System Architecture
- Database Design
- UI/UX Design

Month 3 – Development

- Inventory Module
- User Management
- Alerts & Reports

Month 4 – Testing, Deployment & Training

- Unit Testing

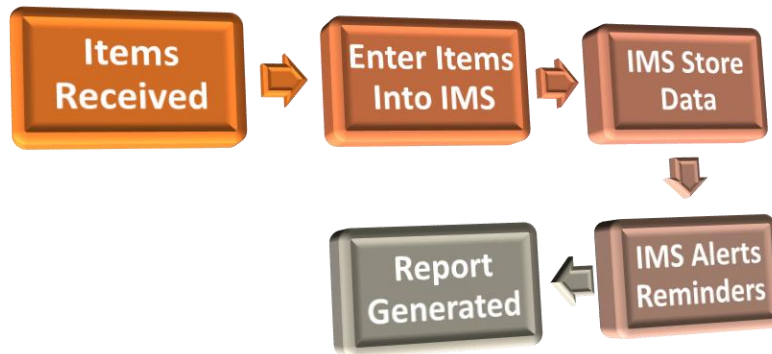
- System Testing
- User Acceptance Testing
- Deployment
- User Training
- Project Closure

9. Possible Risks

- Users may resist changes
- Delays in the schedule
- Requirements might not be clear

10.Exhibits

System Over View



Identifying and Engaging Stakeholders

11.Stakeholder Register

Stakeholder Name	Contact Information	Project Role	Project Requirements	Project Concerns	Influence
MUTANGANA Joseph	+900 588 123 456 josephm@gmail.com	Project Sponsor	System delivered on time and within budget	Cost overruns, delays	Supportive
MUSANGWA Nhsimiye Bennet	+901 722 456 7 bennet@gmail.com	Project Manager	Clear scope, smooth execution	Schedule delays	Supportive
Nshimyumuremyi Ellie	+902 733 987 6 ellie@gmail.com	Inventory Manager	Accurate stock tracking, expiry alerts	Incorrect stock data	Supportive
Shema Abdnur	abdnur@ims.com	Development Team	Clear requirements	Changing requirements	Supportive
Beata Mukamana	+904 729 111 2	Procurement Officer	Low-stock alerts, reorder reports	Late purchase decisions	Supportive

	bmkana@gmail.com				
Eric Bizimana	0783 444 555 ebizi@gmail.com	Finance Officer	Correct inventory valuation reports	Inaccurate costing	Neutral
Samuel Karegeya	5730 999 888 skare@gmail.com	IT Administrator	Secure system, backups	Data loss, system failure	Neutral
Jean Claude	claudej@ims.com	Senior Manger	Project aligns with company goals	Cost, Risk	Supportive

12. Stakeholder Mapping

	High Interest	Low Interest
High Power	Project Sponsor, Project Manager	Senior Management
Low Power	Inventory Manager ,Store Staff , Procurement Officer, Development Team	Finance Officer, IT Administrator

Forming The Team

Team members

Name	Role
MUTANGANA Joseph	Project Sponsor
MUSABWA Nshimiye Bennet	Project Manager
Nshimyumuremyi Ellie	Business Analyst
Shema Abdnur	Lead Developer
Brian	Software Developer
Grace	UI Designer
Michael	Quality Assurance

RACI Chart

Activity	Project Sponsor	Project Manager	Business Analyst	Lead Dev	Soft Dev	UI	QA
Develop Project Charter	I	A	C	I	I	I	I
Define Project Scope	I	A	R	C	I	I	I
Gather Requirements	I	A	R	C	I	I	I
Approve Requirements	I	R	C	I	I	I	I
System Architecture Design	I	I	C	A	C	I	I
UI/UX Design	I	I	C	C	I	A	I
System Development	I	I	C	A	R	I	I
8. System Testing		I	C	C	I	I	A
9. User Acceptance Testing	I	A	R	I	I	I	C
10. Deployment & Project Closure	I	A	C	C	I	I	C

R = Responsible **A** = Accountable **C** = Consult **I** = Informed

TEAM CHARTER

Team Purpose

The purpose of this team is to successfully design, develop, test, and deploy an Inventory Management System (IMS) that improves stock tracking accuracy, reduces shortages and excess inventory, and enhances operational efficiency.

Team Objectives

- Deliver a functional IMS within the approved timeline
- Meet all defined requirements
- Ensure system reliability and usability
- Maintain effective communication among team members

- Deliver quality documentation

Roles and Responsibilities

- **Project Manager** – Oversees planning, scheduling, risk management, stakeholder communication.
- **Business Analyst** – Gathers, documents, and validates requirements.
- **Lead Developer** – Designs system architecture and supervises development.
- **Software Developer** – Implements system features.
- **UI/UX Designer** – Designs system interface and user experience.
- **QA/Test Engineer** – Conducts testing and ensures system quality.

Communication Plan

- Weekly team meeting (every Monday)
- Progress updates via email or WhatsApp group
- Urgent issues communicated immediately
- Formal reports submitted to Project Sponsor bi-weekly

Decision-Making Approach

- Technical decisions → Led by Lead Developer
- Requirement decisions → Led by Business Analyst
- Strategic decisions → Approved by Project Manager
- Major changes → Approved by Project Sponsor

Conflict Resolution

- Issues discussed internally first
- If unresolved → Escalated to Project Manager
- If still unresolved → Sponsor intervention

GROUND RULES

1. All members must attend scheduled meetings.
2. Deadlines must be respected.
3. Members must communicate blockers within 24 hours.

4. Respect all team members' opinions.
5. No task reassignment without PM approval.
6. Documentation must be updated regularly.
7. All work must follow agreed quality standards.

PROJECT SCOPE MANAGEMENT

The Inventory Management System (IMS) will include the following features:

- User authentication and role-based access
- Add new inventory items
- Update item quantities
- Delete items
- Automatic low-stock alerts
- Supplier management module
- Purchase order tracking
- Inventory reporting
- Dashboard overview
- Stock history logs
- Export reports (PDF/Excel)
- User management
- Audit trail
- Data backup functionality

REQUIREMENTS

FR= Functional Requirement **NFR**= Non-Functional Requirement

BR=Business Requirement **TR**= Technical Requirement

Req ID	Requirement Description	Type	Source	Priority	Test Case ID	Status
FR1	The system shall allow users to log in securely.	Functional	Project Sponsor	Must	TC01	Planned
FR2	The system shall allow authorized users to add new inventory items.	Functional	Inventory Manager	Must	TC02	Planned
FR3	The system shall allow updating item quantities.	Functional	Inventory Manager	Must	TC03	Planned

FR4	The system shall allow deletion of obsolete items.	Functional	Inventory Manager	Must	TC04	Planned
FR5	The system shall generate low-stock alerts.	Functional	Inventory Manager	Must	TC05	Planned
FR6	The system shall generate inventory reports.	Functional	Finance Officer	Must	TC06	Planned
FR7	The system shall allow supplier registration.	Functional	Procurement Officer	Should	TC07	Planned
FR8	The system shall record purchase transactions.	Functional	Procurement Officer	Should	TC08	Planned
FR9	The system shall maintain stock history logs.	Functional	IT Administrator	Should	TC09	Planned
FR10	The system shall support user role management.	Functional	Project Sponsor	Must	TC10	Planned
FR11	The system shall allow report export to PDF.	Functional	Finance Officer	Could	TC11	Planned
FR12	The system shall provide dashboard analytics.	Functional	Project Sponsor	Should	TC12	Planned
NFR1	The system shall respond within 3 seconds.	Non-Functional	Project Sponsor	Must	TC13	Planned
NFR2	The system shall be available 95% of operational hours.	Non-Functional	Project Sponsor	Must	TC14	Planned
NFR3	The system shall ensure data encryption.	Non-Functional	IT Administrator	Must	TC15	Planned
NFR4	The system shall support up to 50 concurrent users.	Non-Functional	IT Administrator	Should	TC16	Planned
BR1	The system shall reduce stock shortages by at least 30%.	Business	Project Sponsor	Must	TC17	Planned
BR2	The system shall improve reporting efficiency.	Business	Finance Officer	Should	TC18	Planned
TR1	The system shall use a centralized database.	Technical	Lead Developer	Must	TC19	Planned
TR2	The system shall support automated daily backups.	Technical	IT Administrator	Should	TC20	Planned

MoSCoW PRIORITIZATION

MUST HAVE

- User login and authentication
- Add/Edit/Delete inventory items
- Stock tracking and updates
- Low-stock alerts
- Inventory reports generation
- User role management

SHOULD HAVE

- Dashboard analytics
- Supplier management
- Purchase order tracking
- Stock history logs

COULD HAVE

- Export reports (PDF/Excel)
- Advanced analytics
- Multi-branch support

WON'T HAVE (This Release)

- Mobile application
- AI demand forecasting
- Integration with external External Resources Planning ERP systems

ACCEPTANCE CRITERIA

- Low-stock alert must trigger when stock reaches predefined threshold.
- Reports must generate within 3 seconds.
- User login must authenticate valid credentials only.
- System must prevent unauthorized access.

DELIVERABLES

- Approved Project Charter
- RACI Chart

- Team Charter
- Fully functional IMS software
- User Manual
- Test Report
- Deployment Guide
- Final Project Report

OUT OF SCOPE

- Mobile application development
- Integration with third-party ERP systems
- Multi-language support
- AI-based inventory forecasting
- Cloud hosting infrastructure

CONSTRAINTS

- Limited project timeline
- Limited academic budget
- Team availability constraints
- Technology access limitations

ASSUMPTIONS

- Stakeholders will provide timely feedback.
- Required software tools are available.
- Users have basic computer literacy.
- Internet access is stable during development.

Working Breakdown Structure & Work Packages

1. Project Planning

- 1.1 Develop Project Charter
- 1.2 Identify Stakeholders
- 1.3 Create Project Plan

2. Requirements Analysis

- 2.1 Gather Requirements
- 2.2 Analyze Requirements
- 2.3 Approve Requirements

3. System Design

3.1 Design System Architecture

3.2 Design Database

3.3 Design UI/UX

4. System Development

4.1 Develop Inventory Module

4.2 Develop User Management

4.3 Implement Alerts & Reports

5. System Testing

5.1 Perform Unit Testing

5.2 Perform System Testing

5.3 Conduct User Acceptance Testing

6. Deployment & Closure

6.1 Deploy System

6.2 Train Users

6.3 Project Closure

Working Breakdown Dictionary

1.1	Develop Project Charter	Create initial project document defining objectives and scope	3 days
1.2	Identify Stakeholders	Identify all stakeholders involved in the project	2 day
1.3	Create Project Plan	Develop overall project plan including schedule and resources	5 days
2.1	Gather Requirements	Collect system requirements from stakeholders	7 days
2.2	Analyze Requirements	Review and refine collected requirements	5 days
2.3	Approve Requirements	Validate and approve requirements with stakeholders	3 days
3.1	Design System Architecture	Define system structure and components	6 days
3.2	Design Database	Create database schema for inventory data	5 days
3.3	Design UI/UX	Design system interface and user experience	6 days
4.1	Develop Inventory Module	Implement core inventory management features	12 days

4.2	Develop User Management	Implement user roles and authentication	8 days
4.3	Implement Alerts & Reports	Develop alerts and reporting features	10 days
5.1	Perform Unit Testing	Test individual components of the system	4 days
5.2	Perform System Testing	Test complete system functionality	6 days
5.3	Conduct User Acceptance Testing	Validate system with end users	4 days
6.1	Deploy System	Install and configure system for use	3 days
6.2	Train Users	Train users on how to use the system	5 days
6.3	Project Closure	Finalize documentation and close project	2 day

Activity Dependencies

Activity	Depends On
Requirements Analysis	Planning
System Design	Requirements
Development	Design
Testing	Development
Deployment	Testing

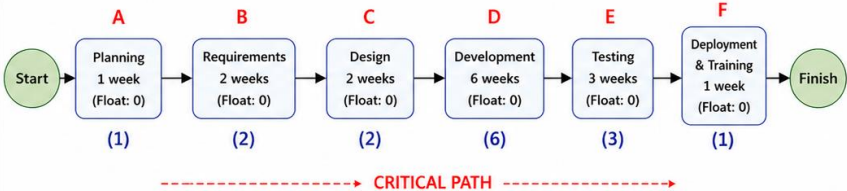
Time Estimation

Phase	Duration
Planning	1 weeks
Requirements	2 weeks
Design	2 weeks
Development	6 weeks
Testing	3 weeks
Deployment	1 weeks

PROJECT NETWORK DIAGRAM (CPM)

3-4 Month Project Timeline (16 Weeks)

ID	Activity	Duration (Weeks)	Predecessor
A	Planning	1	-
B	Requirements	2	A
C	Design	2	B
D	Development	6	C
E	Testing	3	D
F	Deployment & Training	1	E



EARLIEST, LATEST, FLOAT TIMES (WEEKS)						
ID	Activity	Duration (Weeks)	Earliest		Latest	
			ES	EF	LS	LF
A	Planning	1	0	1	0	1
B	Requirements	2	1	3	1	3
C	Design	2	3	5	3	5
D	Development	6	5	11	5	11
E	Testing	3	11	14	11	14
F	Deployment & Training	1	14	15	14	15

All activities have zero float, therefore the critical path is:
A → B → C → D → E → F
(Total duration = 15 weeks)

SUMMARY	
• Critical Path Duration	: 15 weeks
• Total Project Timeline	: 16 weeks (4 months)
• Buffer / Contingency	: 1 week (~6% of total duration)
Buffer of 1 week is kept at the end to handle risks, changes, and unforeseen delays.	

3-4 MONTH PROJECT TIMELINE (16 WEEKS)																
	Month 1 (Weeks 1-4)				Month 2 (Weeks 5-8)				Month 3 (Weeks 9-12)				Month 4 (Weeks 13-16)			
Week	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Planning																
Requirements																
Design																
Development																
Testing																
Deployment & Training																
Buffer / Contingency																

NOTES	
• The critical path determines the minimum time to complete the project.	
• Any delay in activities on the critical path will delay the project completion.	
• A 1-week buffer at the end is included for risk, changes, rework, and unforeseen delays.	
• Deployment & Training includes system deployment, user training, and handover.	

Resource Calendar

Resource	Type	Availability	Purpose
Project Manager	Human	Full project duration	Project planning, coordination, and control
Business Analyst	Human	Planning & Requirements phase	Gathering and analyzing requirements
Lead Developer	Human	Design, Development & Deployment phases	System architecture and technical supervision
Software Developer	Human	Development phase	System implementation
UI/UX Designer	Human	Design phase	Interface and user experience design
QA/Test Engineer	Human	Testing phase and Development phase	System testing and quality assurance
Laptops	Technical	Full project duration	Development, testing, and documentation
Internet Connection	Technical	Full project duration	Communication and system access
Development Software (IDE)	Technical	Design & Development phases	Coding the system
Database System	Technical	Development & Testing phases	Data storage and management
Testing Tools	Technical	Testing phase	System testing activities

Risks

Negative Risks

Risk	Response	Justification
Requirement misunderstanding	Mitigate	Regular meetings reduce errors
Development delay	Mitigate	Proper scheduling helps control time
System bugs	Mitigate	Testing reduces defects
Poor internet connection	Accept	Cannot fully control
Stakeholder unavailability	Mitigate	Early communication planning

Positive Risks

Risk	Response	Justification
Faster development	Exploit	Deliver earlier
Skilled developer performance	Enhance	Assign critical tasks
Good team collaboration	Enhance	Improves productivity
Early completion of testing	Exploit	Move to deployment faster
Availability of better tools	Enhance	Improves efficiency